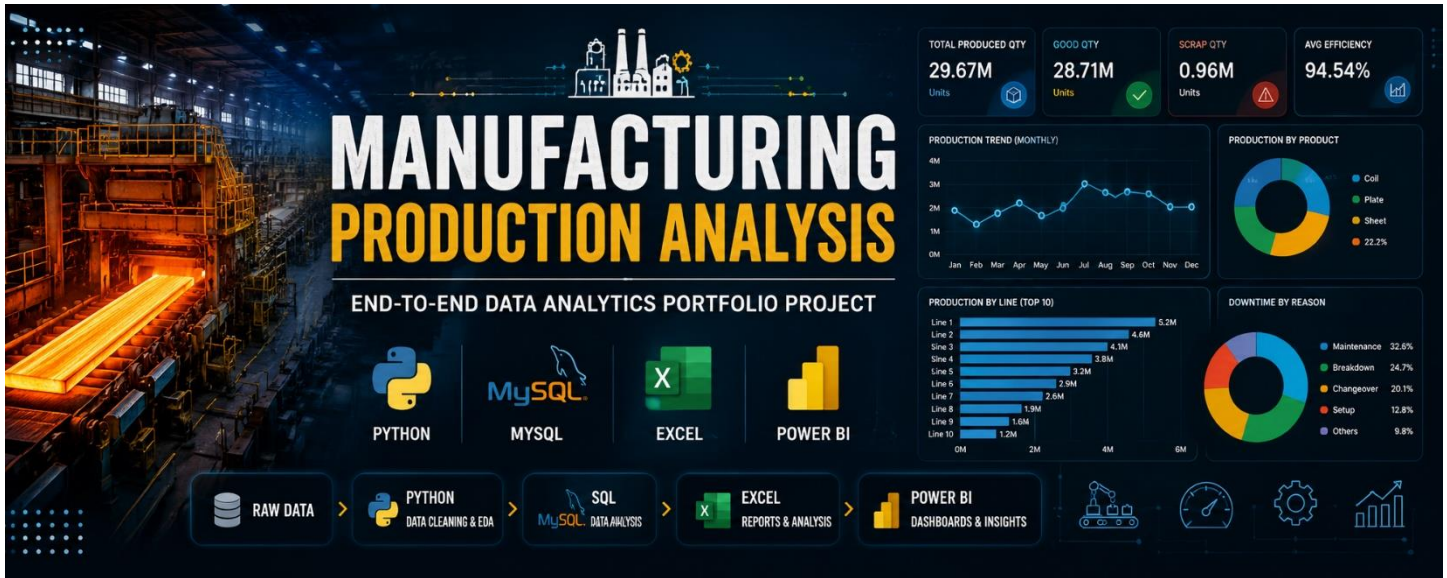




Manufacturing Production Analysis

End-to-End Data Analytics Portfolio Project

Domain:

Manufacturing / Steel Production

Prepared By:

Harshal Dabhire

Technology Stack:

Python | MySQL | Microsoft Excel | Power BI

Technical Project Report

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Project Overview

1.1 Project Summary

This project presents an end-to-end data analytics workflow developed for a manufacturing production environment. Using a synthetic steel production dataset, the project demonstrates the complete analytical lifecycle, beginning with raw data generation and progressing through data preparation, business analysis, reporting, and dashboard development.

The workflow integrates Python for data cleaning and exploratory analysis, MySQL for business-oriented querying, Microsoft Excel for KPI reporting, and Power BI for interactive dashboard creation. Each stage produces outputs that support the next phase of analysis, resulting in a structured and reproducible analytics pipeline.

1.2 Project Scope

The scope of the project includes preparing production data for analysis, validating data quality, engineering analytical features, performing business analysis using SQL, creating KPI-based reports in Microsoft Excel, and developing interactive dashboards in Power BI. The project focuses on descriptive analytics and operational reporting using historical production data.

1.3 End-to-End Analytics Approach

The project follows a sequential analytics workflow in which raw manufacturing data is transformed into business-ready information through multiple analytical stages. The cleaned dataset generated in Python serves as the common data source for SQL analysis, Excel reporting, and Power BI dashboards, ensuring consistency across all reporting outputs.

2. Business Problem

2.1 Manufacturing Data Challenge

Manufacturing organizations generate large volumes of operational data from production activities, including planned production, actual output, machine performance, downtime, and product quality. Although this data contains valuable operational information, it is often distributed across multiple records and requires structured analysis before meaningful insights can be obtained.

Without a systematic analytical process, it becomes difficult to evaluate production performance, identify operational inefficiencies, monitor equipment utilization, or measure production losses. As a result, decision-making may rely on isolated observations rather than consistent, data-driven analysis.

2.2 Need for Production Analytics

Production analytics transforms raw manufacturing data into actionable business information by measuring operational performance through key performance indicators (KPIs) and trend analysis. By integrating data preparation, business analysis, and visualization, organizations can better understand production efficiency, downtime patterns, product quality, and overall manufacturing performance.

This project demonstrates how an end-to-end analytics workflow can support descriptive analysis and business reporting using manufacturing production data.

3. Project Objectives

3.1 Data Preparation Objectives

The initial objective was to prepare a reliable analytical dataset by improving data quality through cleaning, validation, and feature engineering. This stage ensured that the dataset was consistent, standardized, and suitable for downstream analysis.

3.2 Analysis Objectives

The project aimed to analyze production performance using business-oriented metrics across multiple operational dimensions, including production lines, shifts, products, machines, operators, and downtime categories. SQL queries were used to summarize operational data and generate business insights.

3.3 Reporting Objectives

The final objective was to present analytical results through structured reports and interactive dashboards. Microsoft Excel was used for KPI reporting and pivot-based analysis, while Power BI provided interactive visualizations that support business monitoring and performance evaluation.

4. Project Workflow and Data Pipeline

4.1 End-to-End Workflow

The project follows a sequential analytics workflow in which each stage builds upon the output generated by the previous stage. The process begins with a generated manufacturing dataset, followed by data preparation in Python, business analysis in MySQL, reporting in Microsoft Excel, and interactive dashboard development in Power BI.

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4.2 Data Processing Pipeline

The analytical workflow consists of the following stages:

Dataset Generation: Creation of the manufacturing production dataset.

Python Processing: Data cleaning, validation, feature engineering, and exploratory data analysis.

Processed Dataset: Export of the cleaned dataset (manufacturing_cleaned.csv) for downstream analysis.

MySQL Analysis: Business-oriented querying and KPI analysis.

Microsoft Excel Reporting: KPI dashboards, pivot analysis, and business reporting.

Power BI Dashboard: Interactive dashboards with data modeling, DAX measures, and business visualizations.

This structured pipeline ensures consistency across all analytical stages while maintaining a single processed dataset as the foundation for reporting and visualization.

5. Dataset Overview

5.1 Dataset Source

The project uses a synthetically generated manufacturing production dataset designed to simulate operational data commonly found in a steel manufacturing environment. The dataset serves as the primary input for data preparation, business analysis, and reporting throughout the project.

5.2 Dataset Characteristics

The dataset contains production records representing manufacturing operations across multiple production lines, shifts, machines, operators, and products. It includes quantitative production metrics, operational timestamps, and downtime information required for descriptive analytics and performance reporting.

The primary analytical attributes include production quantities, scrap quantities, downtime duration, efficiency metrics, production dates, product information, and operational identifiers.

5.3 Data Quality Assessment

Before analysis, the dataset was evaluated for common data quality issues. The assessment included identifying missing values, duplicate records, invalid production values, inconsistent text formatting, and incorrect data types. Appropriate cleaning and validation procedures were applied to improve the overall quality and consistency of the analytical dataset.

5.4 Engineered Features

The following features were created during the Python data preparation stage to support business analysis and reporting:

Feature	Description
Good_Qty	Quantity of successfully produced units after accounting for scrap.
Production_Loss	Difference between planned production and actual production.
Downtime_Hours	Downtime converted from minutes to hours for reporting purposes.
Month	Month extracted from the production date for monthly analysis.
Quarter	Quarter derived from the production date for period-based reporting.
Week	Week number extracted for weekly trend analysis.
Day_Name	Day of the week derived from the production date for daily performance analysis.
Efficiency_Category	Classification of production efficiency into predefined performance groups.

6. Python Data Preparation & Analysis

Python was used as the primary environment for preparing the manufacturing dataset for analysis. This stage focused on improving data quality, validating production records, engineering analytical features, and performing exploratory data analysis (EDA). The resulting cleaned dataset served as the common data source for the SQL, Microsoft Excel, and Power BI stages of the project.

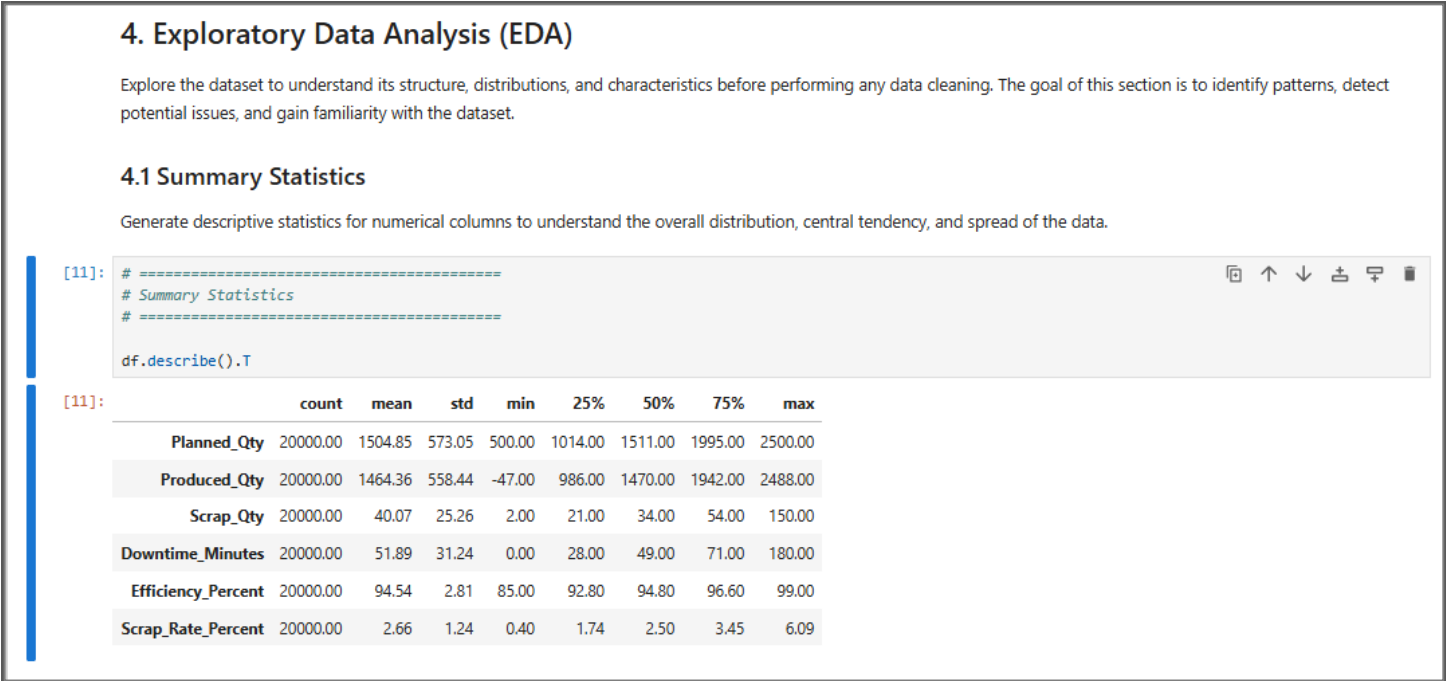
6.1 Data Loading and Exploration

The manufacturing production dataset (manufacturing_data.csv) was imported into Python using the Pandas library. An initial assessment was performed to understand the dataset structure, verify data types, inspect column information, and identify potential data quality issues before preprocessing.

The exploratory phase included:

- Inspecting dataset dimensions and column information.
- Reviewing data types and summary statistics.
- Identifying missing values and duplicate records.
- Examining the distribution of key production metrics.
- Understanding production, scrap, downtime, and efficiency variables.

These activities established a baseline for the data preparation process.



[python.png]

Figure 6.1. Python notebook illustrating data preparation and exploratory data analysis.

6.2 Data Cleaning Process

A structured data cleaning workflow was implemented to improve dataset consistency and analytical reliability. The following preprocessing activities were performed:

- Missing values were identified and handled appropriately.
- Duplicate records were detected and removed.
- Invalid production records were identified through validation checks and excluded from the cleaned dataset.
- Text fields were standardized to maintain consistent formatting.
- Data types were converted to appropriate formats for numerical and date-based analysis.

These steps produced a reliable dataset suitable for downstream business analysis and reporting.

6.3 Feature Engineering

To enhance analytical capabilities, several derived features were created from the original production data. These engineered attributes supported operational analysis, KPI calculations, and time-based reporting across subsequent project stages.

Engineered Feature	Purpose
Good_Qty	Calculates the quantity of successfully produced units after accounting for scrap.
Production_Loss	Measures the difference between planned and actual production output.
Downtime_Hours	Converts downtime from minutes to hours for standardized reporting.
Month	Enables monthly production analysis.
Quarter	Supports quarterly performance reporting.
Week	Enables weekly trend analysis.
Day_Name	Supports day-wise operational analysis.
Efficiency_Category	Classifies production efficiency into performance groups for comparative analysis.

These derived features simplified business analysis and improved reporting across SQL queries, Excel dashboards, and Power BI visualizations.

6.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand production behavior and identify patterns within the manufacturing dataset. Statistical summaries and visualizations were used to examine operational performance and verify data consistency.

The analysis focused on:

- Production quantity distribution.
- Planned versus actual production.
- Scrap quantity analysis.
- Downtime analysis.
- Production efficiency evaluation.
- Time-based production trends.

The findings from EDA guided the validation process and confirmed that the cleaned dataset was suitable for business reporting.

6.5 Data Validation and Export

Following preprocessing and exploratory analysis, the dataset underwent validation to verify consistency and analytical readiness. Validation checks ensured that production values satisfied defined business rules and that engineered features were generated correctly.

After successful validation, the processed dataset was exported as **manufacturing_cleaned.csv**, providing a standardized dataset for all subsequent analytical stages of the project.

This cleaned dataset became the single source of truth for SQL analysis, Microsoft Excel reporting, and Power BI dashboard development, ensuring consistency across all project deliverables.

End of Python Stage

The Python stage established a structured data preparation pipeline by transforming raw manufacturing records into a clean, validated, and analysis-ready dataset. This foundation enabled consistent business analysis and visualization throughout the remainder of the project.

7. SQL Business Analysis

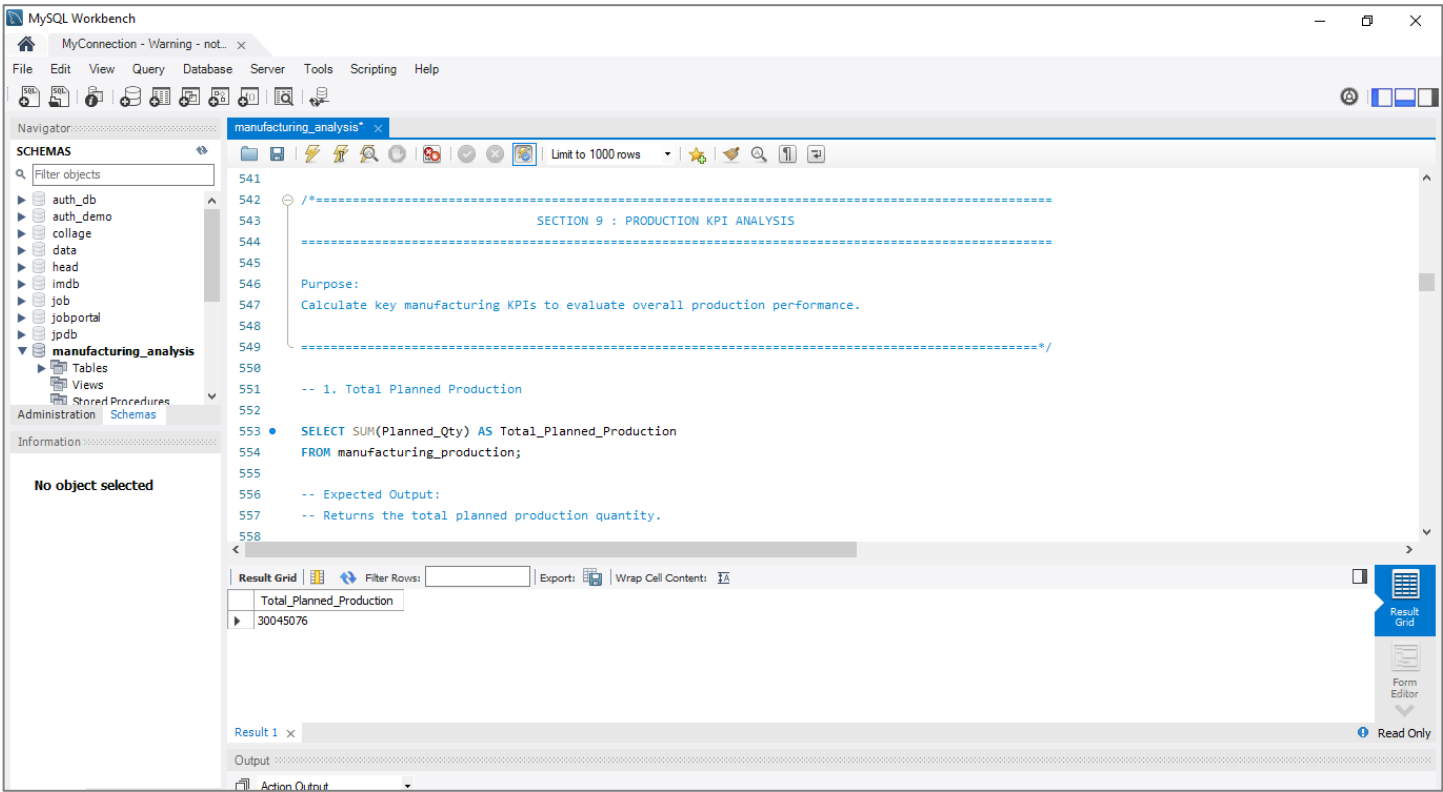
After preparing the dataset in Python, the cleaned dataset (manufacturing_cleaned.csv) was imported into MySQL for business-oriented analysis. SQL was used to transform processed production data into meaningful operational information through filtering, aggregation, and analytical queries.

The SQL stage focused on evaluating manufacturing performance across multiple operational dimensions, supporting KPI calculation and business reporting.

7.1 Database Setup

The cleaned dataset was imported into a MySQL database, where the production records were organized into a structured relational table. Appropriate data types were assigned to support efficient querying, filtering, aggregation, and analytical operations.

The database environment served as the foundation for all business analysis performed during this stage of the project.



sql_analysis.png

Figure 7.1. SQL business analysis using MySQL queries.

7.2 Basic Data Analysis

Initial SQL queries were used to explore and summarize the production dataset. Common SQL operations helped organize and interpret manufacturing data before performing advanced analysis.

The analysis included:

- Data filtering using WHERE clauses.
- Sorting production records.
- Aggregate calculations using SQL functions.
- Grouping records using GROUP BY.
- Applying conditions with HAVING.

These operations provided summarized production metrics across different operational categories.

7.3 KPI Analysis

SQL queries were developed to calculate key manufacturing performance indicators that support operational monitoring and business reporting.

The analysis focused on:

- Planned production.
- Actual production.
- Good production quantity.
- Production loss.
- Scrap quantity.
- Average production efficiency.
- Average downtime.
- Scrap rate.

These KPIs provide a quantitative overview of manufacturing performance and form the basis for subsequent reporting in Microsoft Excel and Power BI.

7.4 Operational Analysis

Manufacturing performance was analyzed across multiple operational dimensions to better understand production behavior and identify performance variations.

The analysis included:

- Production line analysis.
- Shift-wise performance analysis.
- Product-wise production analysis.
- Machine performance analysis.
- Operator performance analysis.

This multidimensional approach enabled comparisons across operational categories and supported descriptive business analysis.

7.5 Time and Downtime Analysis

Time-based SQL queries were used to evaluate production performance across different reporting periods and to examine operational downtime.

The analysis included:

- Monthly production trends.
- Weekly performance analysis.
- Downtime by operational reason.
- Production performance over time.

These analyses supported a better understanding of manufacturing trends and operational interruptions.

7.6 Advanced SQL Techniques

To improve query efficiency and demonstrate advanced SQL capabilities, several analytical SQL features were incorporated during the project.

The implementation included:

- Window Functions for analytical calculations.
- Common Table Expressions (CTEs) for modular query design.
- Views for reusable business reporting.

These techniques improved query readability, organization, and maintainability while supporting more structured business analysis.

End of SQL Stage

The SQL stage transformed the cleaned manufacturing dataset into meaningful business information through structured querying, KPI calculation, and operational analysis. The generated insights provided the analytical foundation for the reporting and dashboard development stages completed in Microsoft Excel and Power BI.

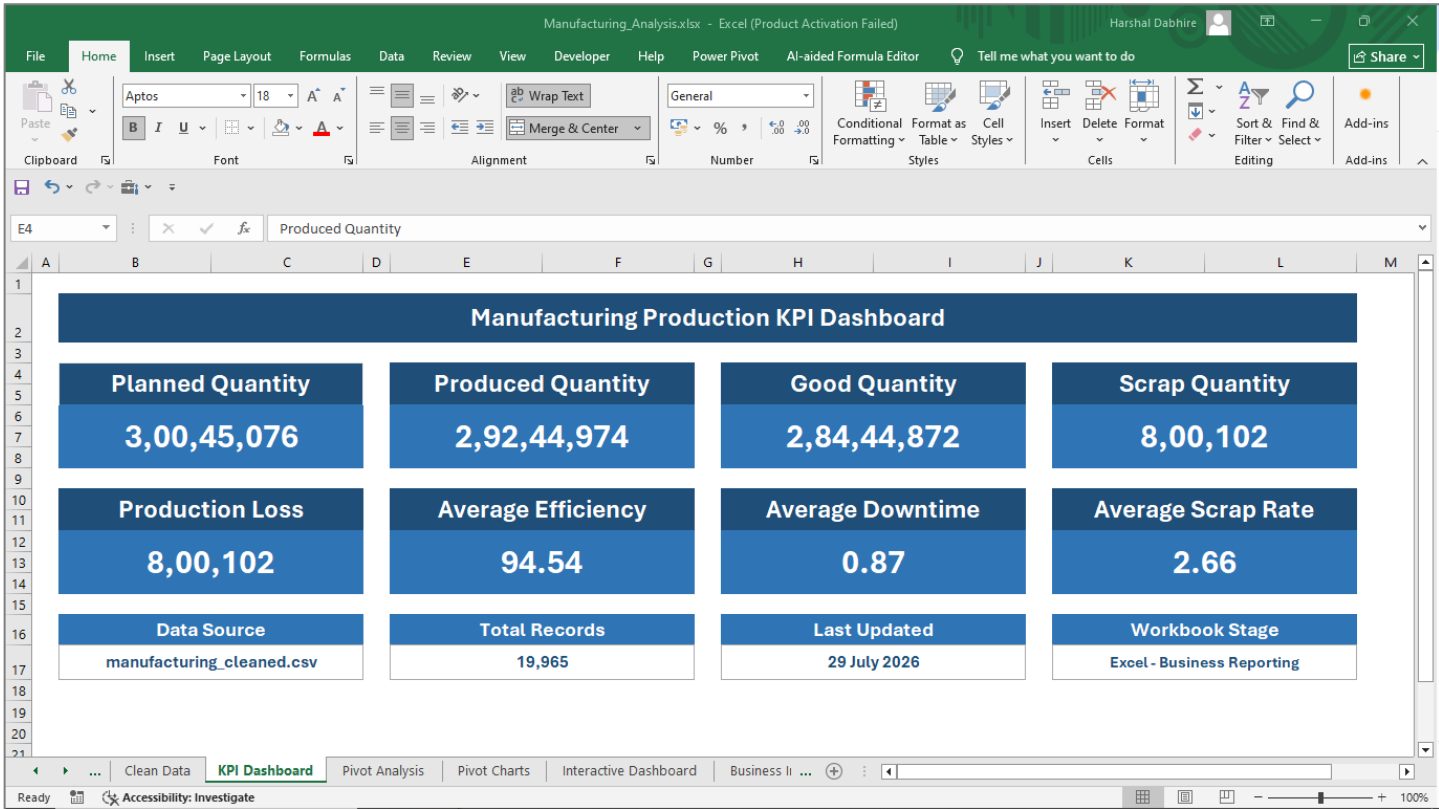
8. Microsoft Excel Reporting

After completing the SQL analysis, the processed manufacturing dataset was analyzed in Microsoft Excel to create business reports and interactive dashboards. Excel was used to summarize production data, calculate key performance indicators (KPIs), and present operational information through pivot tables, charts, and dashboard components.

This stage focused on transforming analytical results into a business-friendly reporting format suitable for monitoring manufacturing performance.

8.1 Data Preparation

The cleaned manufacturing dataset was imported into Microsoft Excel and organized into structured worksheets for reporting and analysis. The workbook was designed to separate raw data, cleaned data, pivot analysis, and dashboard components, ensuring an organized reporting workflow.



excel_dashboard.png

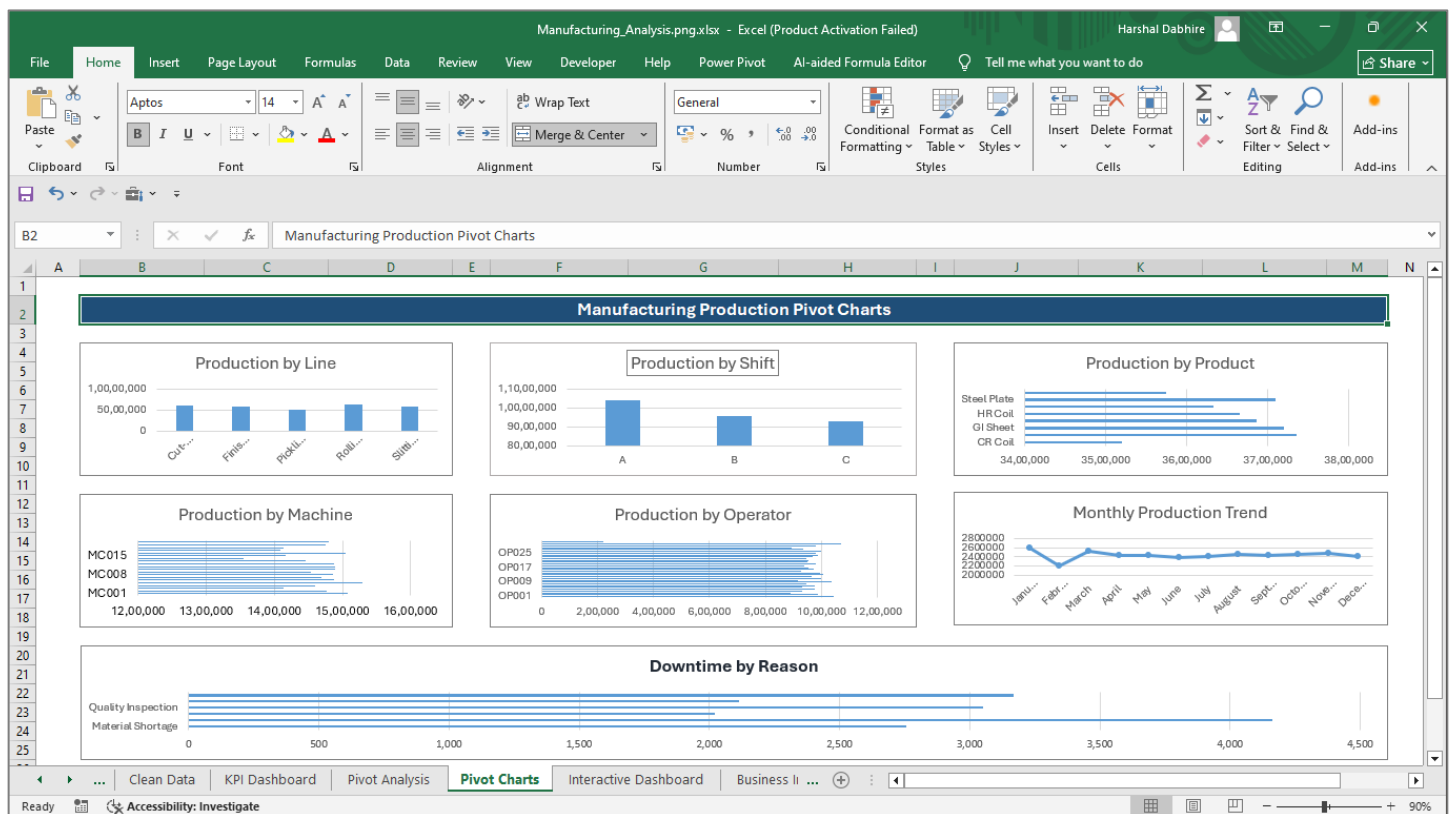
Figure 8.1. Microsoft Excel KPI dashboard summarizing manufacturing performance.

8.2 KPI Dashboard

A KPI dashboard was developed to provide a consolidated view of key manufacturing performance metrics. The dashboard summarizes production performance using calculated indicators that support quick business evaluation.

The primary KPIs include:

- Planned Production Quantity
- Produced Quantity
- Good Quantity
- Scrap Quantity
- Production Loss
- Average Efficiency
- Average Downtime
- Average Scrap Rate



excel_charts.png

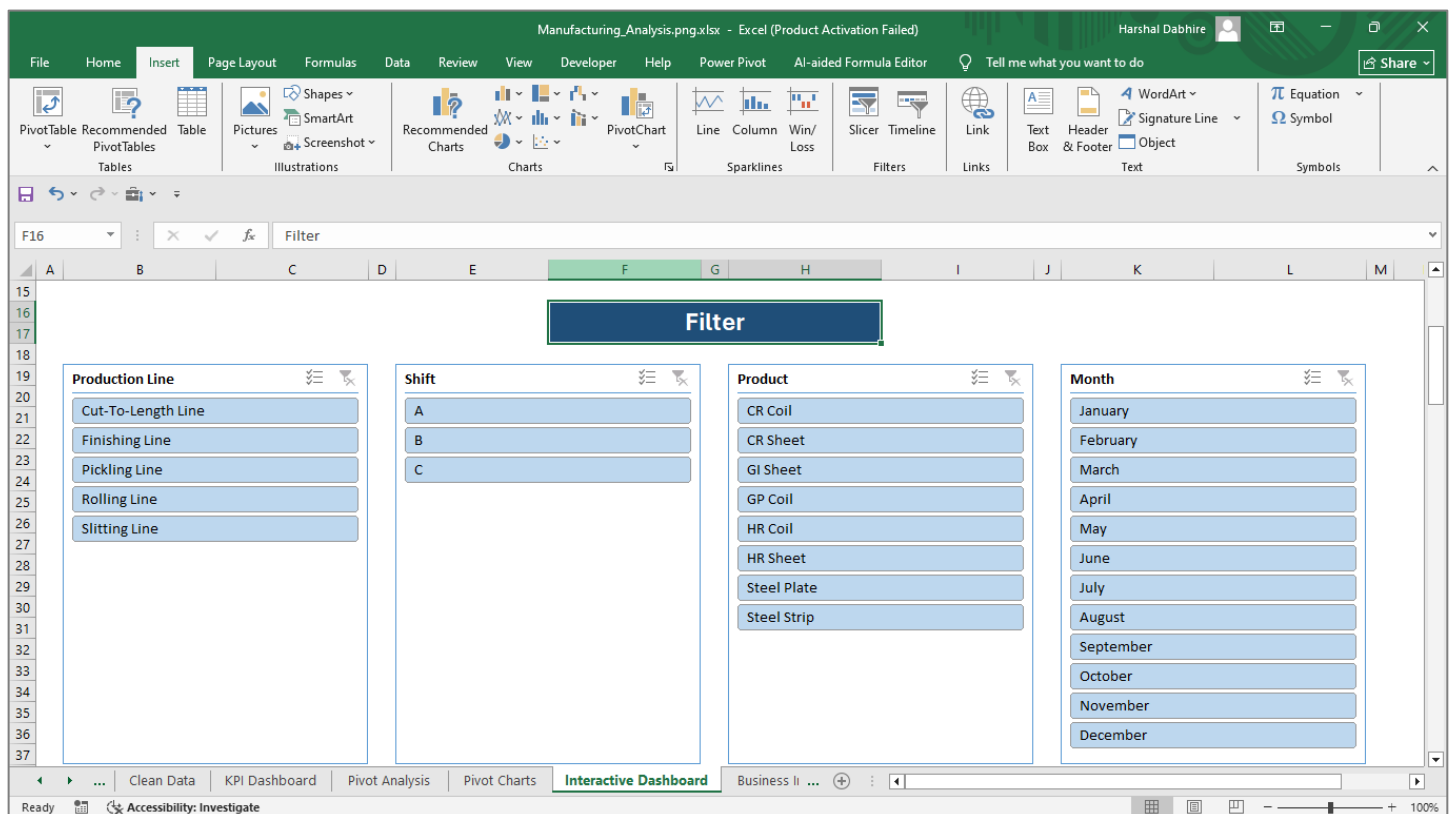
Figure 8.2. Pivot charts used for manufacturing performance analysis.

8.3 Pivot Analysis

Pivot Tables were used to summarize manufacturing data across different operational dimensions, enabling efficient comparison of production performance and supporting descriptive business analysis.

The analysis included:

- Production Line Analysis
- Shift-wise Analysis
- Product-wise Analysis
- Machine Performance
- Operator Performance
- Monthly Production Trends
- Downtime Analysis



excel_filters.png

Figure 8.3. Interactive filters and slicers within the Microsoft Excel dashboard.

8.4 Pivot Charts and Visualization

Pivot Charts were created to transform summarized data into graphical representations, making production trends and operational performance easier to interpret.

The visualizations support comparison across multiple manufacturing categories while improving the readability of business reports.

8.5 Interactive Dashboard

Interactive filters and slicers were incorporated into the Excel dashboard to improve report usability and allow users to explore manufacturing data dynamically. These controls enable focused analysis across different operational categories without modifying the underlying dataset.

End of Excel Stage

The Microsoft Excel stage converted analytical results into structured business reports through KPI dashboards, Pivot Tables, Pivot Charts, and interactive filtering. These reporting components provide an accessible view of manufacturing performance and complement the SQL analysis with user-friendly visual summaries.

9.Power BI Dashboard

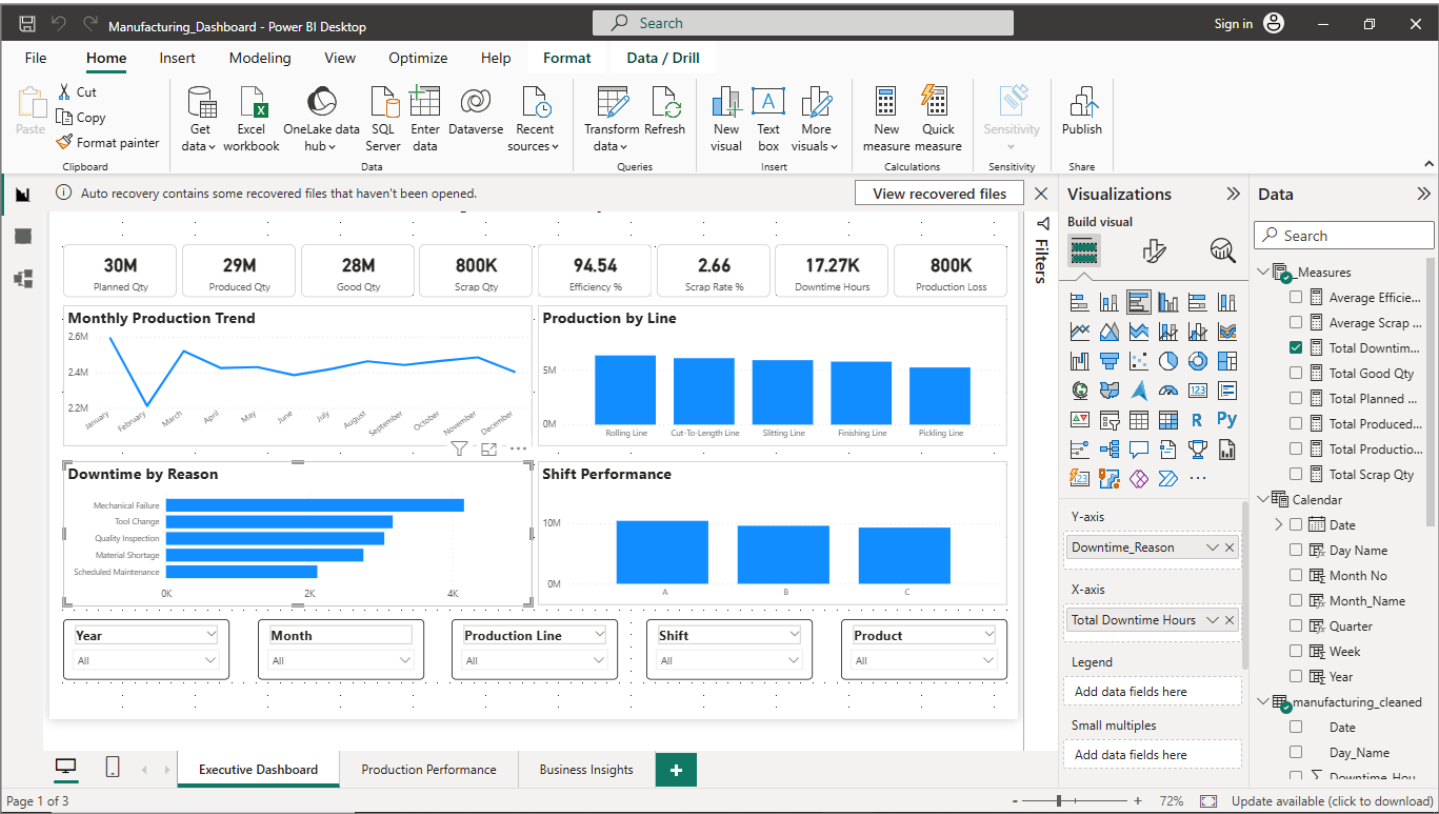
Power BI was used to transform the processed manufacturing dataset into an interactive business intelligence solution. The dashboard combines key performance indicators, operational visualizations, and interactive filtering to support efficient monitoring of manufacturing performance.

The report was developed using the cleaned dataset generated during the Python stage, ensuring consistency across all analytical and reporting components.

9.1 Data Modeling

The cleaned manufacturing dataset was imported into Power BI and prepared for dashboard development. Data modeling involved organizing the dataset and establishing the required relationships to support accurate calculations and interactive reporting.

The prepared data model provided the foundation for all dashboard visualizations and analytical measures.



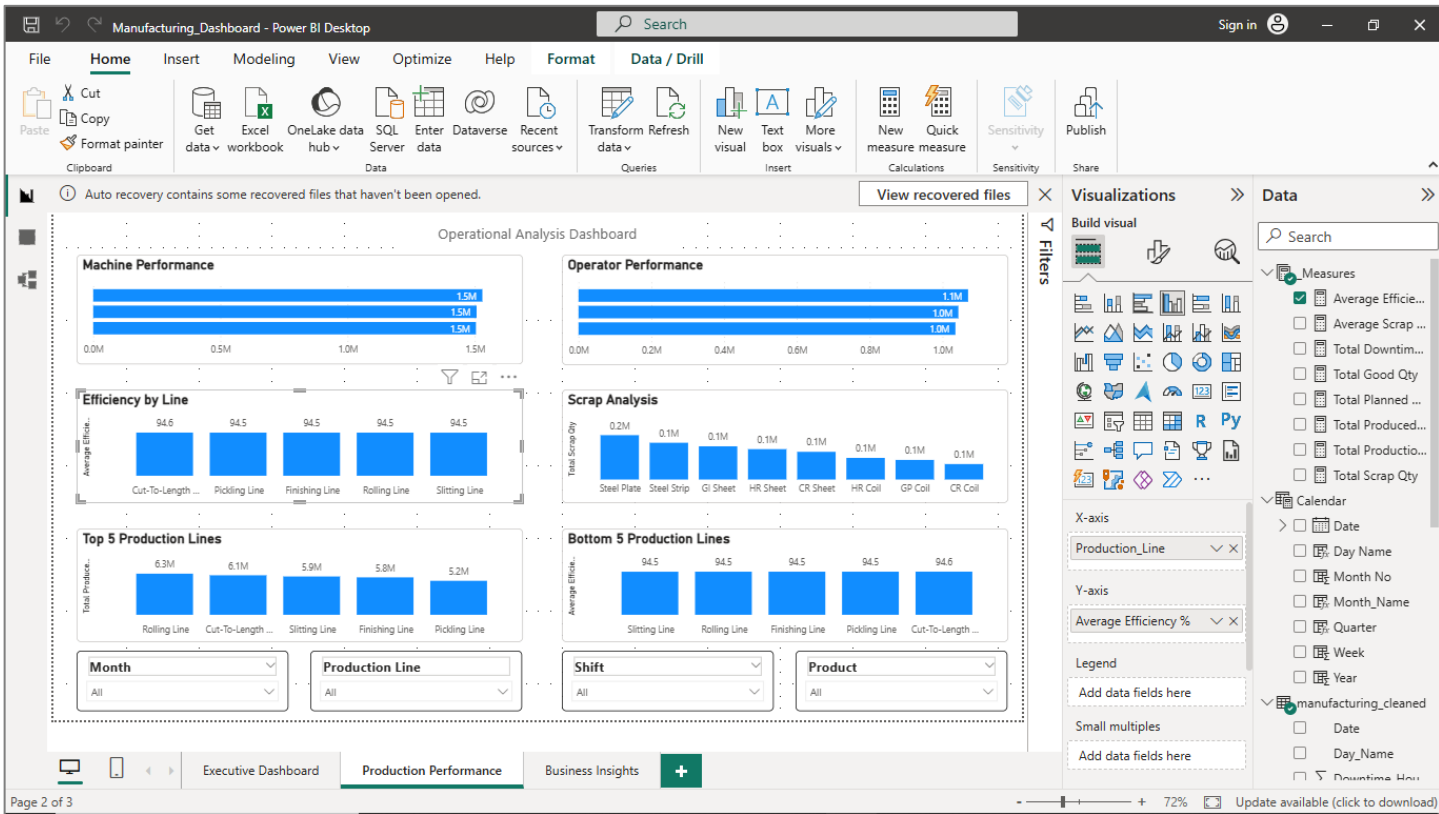
powerbi_page1.png

Figure 9.1. Executive dashboard providing an overview of manufacturing KPIs and operational performance.

9.2 DAX Measures

Data Analysis Expressions (DAX) were used to calculate business metrics displayed throughout the dashboard. These measures summarize production performance and support dynamic calculations based on user selections and report filters.

The implemented measures include production quantities, efficiency metrics, production loss, downtime, and quality-related KPIs used across multiple report pages.



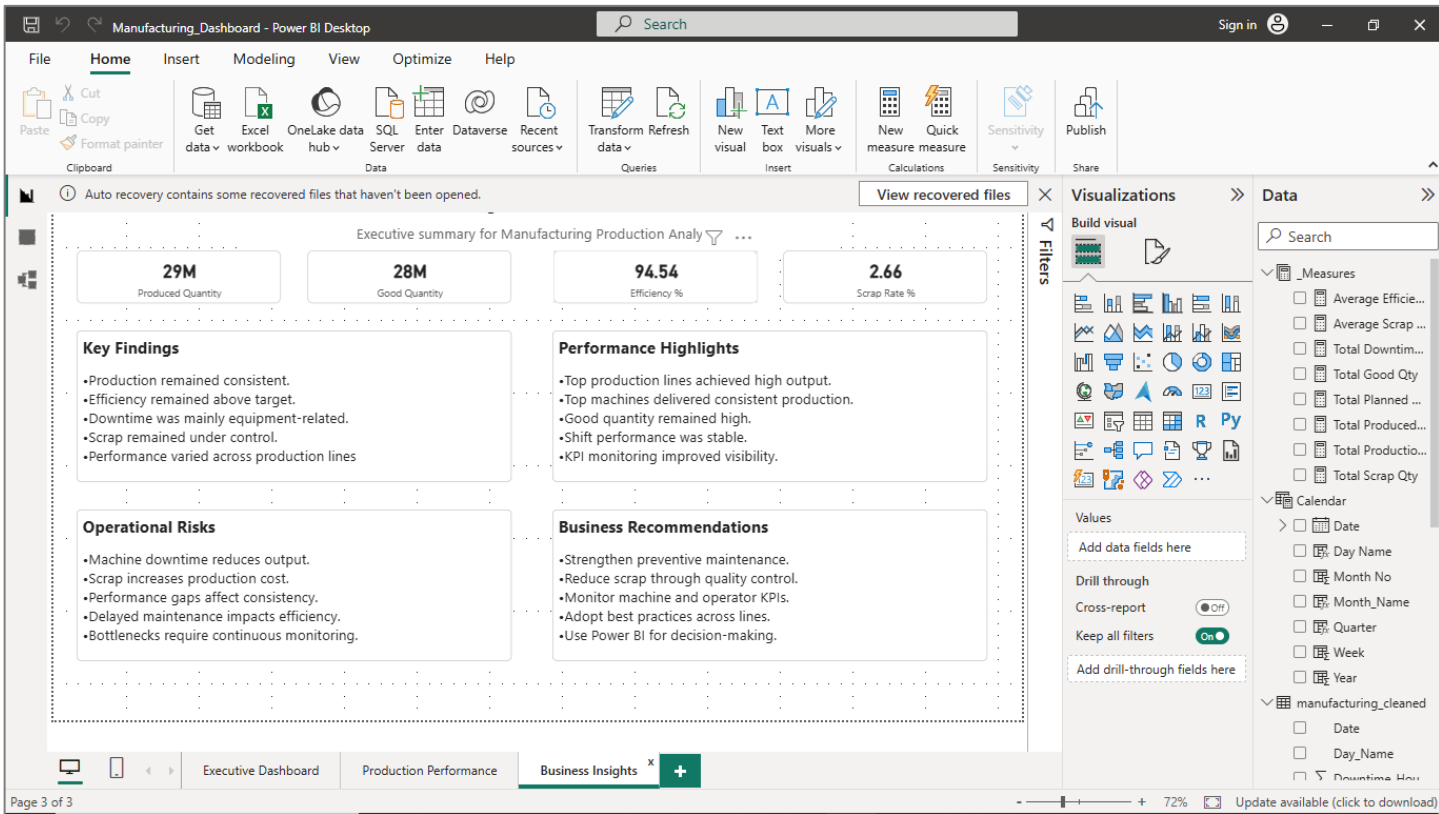
powerbi_page2.png

Figure 9.2. Operational analysis dashboard presenting manufacturing performance across multiple business dimensions.

9.3 Interactive Dashboard Design

The dashboard was designed to present manufacturing information through an intuitive layout that combines KPI cards, charts, and slicers. Interactive filters allow users to explore production data across different operational categories while maintaining a consistent reporting experience.

The dashboard supports dynamic analysis without modifying the underlying dataset, enabling efficient business reporting and performance monitoring.



powerbi_page3.png

Figure 9.3. Detailed analytical dashboard supporting operational insights and business reporting.

9.4 Dashboard Pages

Page 1 – Executive Overview

The first dashboard page provides a high-level summary of manufacturing performance through KPI cards and visual summaries. It enables users to quickly evaluate production efficiency, production output, downtime, and quality-related metrics.

Page 2 – Operational Analysis

The second dashboard page focuses on operational performance by presenting production data across multiple business dimensions such as production lines, shifts, products, machines, and operators. This view supports comparative analysis and operational monitoring.

Detailed Insights

The third dashboard page provides additional analytical views that support deeper exploration of manufacturing performance. It complements the executive overview by presenting detailed operational trends and business insights through interactive visualizations.

End of Power BI Stage

The Power BI stage integrates data modeling, DAX calculations, and interactive visualizations into a business intelligence solution that supports descriptive analysis and operational reporting. The dashboard provides a centralized view of manufacturing performance while enabling users to explore data through interactive filters and multiple analytical perspectives.

10. Key Business Insights

This section should present the **analytical findings**, not describe charts or dashboards again. Since this is a portfolio project using a synthetic dataset, the insights should be **generic but technically valid**, avoiding unsupported claims.

10.1 Production Performance Insights

The analysis of production records indicates that comparing planned production with actual production provides a clear measure of operational performance. Tracking production loss alongside good production quantity enables better evaluation of manufacturing efficiency and highlights areas where production targets are not fully achieved.

10.2 Efficiency and Quality Insights

Production efficiency and scrap rate are key indicators of manufacturing quality. Evaluating these metrics together provides a balanced view of operational performance, allowing quality issues to be identified alongside production output. Categorizing efficiency levels further simplifies performance monitoring and comparison across different operational groups.

10.3 Downtime Insights

Downtime analysis helps identify periods during which production is interrupted and supports the evaluation of operational reliability. By examining downtime duration and associated reasons, manufacturing managers can better understand operational disruptions and prioritize improvement efforts where recurring downtime patterns exist.

10.4 Operational Observations

The combined use of Python, SQL, Microsoft Excel, and Power BI demonstrates how manufacturing data can be transformed into structured business information. The integrated workflow supports KPI monitoring, trend analysis, and operational reporting while providing a consistent analytical foundation from data preparation through dashboard development.

11. Project Outcomes

11.1 Technical Outcomes

The project successfully demonstrates an end-to-end data analytics workflow using a manufacturing production dataset. Throughout the project, raw production data was transformed into a clean and analysis-ready dataset through structured data preparation, validation, and feature engineering. The processed data was subsequently analyzed using SQL, reported through Microsoft Excel, and visualized using Power BI.

The project also demonstrates practical application of Python, MySQL, Microsoft Excel, and Power BI within a single analytical workflow, illustrating how multiple technologies can be integrated to support data-driven decision-making.

11.2 Business Outcomes

The analytical workflow provides a structured approach for monitoring manufacturing operations through key performance indicators, production analysis, and interactive dashboards. The reporting components support evaluation of production output, operational efficiency, downtime, and quality-related metrics.

Although developed as a portfolio project using a synthetic dataset, the overall workflow reflects common analytical practices used in manufacturing reporting and business intelligence.

12. Technology Stack

The project integrates multiple analytical tools, each serving a specific role within the overall workflow.

Technology	Purpose
Python	Data cleaning, validation, feature engineering, and exploratory data analysis
Pandas	Data manipulation and preprocessing
NumPy	Numerical operations and calculations
Matplotlib	Data visualization during exploratory analysis
MySQL	Business analysis using SQL queries and KPI calculations
Microsoft Excel	KPI reporting, Pivot Tables, Pivot Charts, and interactive dashboards
Power BI	Data modeling, DAX measures, and interactive business intelligence dashboards
Git	Version control during project development
GitHub	Project repository and portfolio presentation

13. Project Limitations

13.1 Dataset Limitations

The project uses a synthetically generated manufacturing dataset created for educational and portfolio purposes. While the dataset represents realistic manufacturing scenarios, it does not originate from a live production environment.

13.2 Analytical Limitations

The analysis focuses on descriptive analytics using historical production data. Predictive modeling, forecasting, anomaly detection, and machine learning techniques are outside the scope of this project.

13.3 Deployment Limitations

The project was developed as a standalone analytics portfolio and does not include automated data pipelines, scheduled data refresh, cloud deployment, or integration with live manufacturing systems.

14. Conclusion

14.1 Final Summary

This project demonstrates a complete end-to-end data analytics workflow for manufacturing production analysis, beginning with raw dataset generation and progressing through data preparation, business analysis, reporting, and dashboard development. Each stage contributes to transforming operational data into structured business information suitable for analysis and decision support.

By integrating Python, MySQL, Microsoft Excel, and Power BI within a single workflow, the project showcases practical data analytics skills across multiple technologies while maintaining a consistent analytical dataset throughout the reporting process. The completed project serves as a comprehensive portfolio example of descriptive analytics and business intelligence applied to a manufacturing production scenario.
